

**FRESHROUTE - SMART FRUIT SUPPLY CHAIN  
SYSTEM TO REDUCE WASTE AND ELIMINATE  
MIDDLEMEN IN SRI LANKA**

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Project Proposal Report

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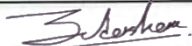
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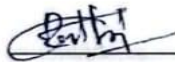
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## DECLARATION

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## ABSTRACT

Fruit quality plays an important role in the fresh fruit supply chain, as both farmers and buyers expect transparency and fairness in trade. However, the current process of evaluating fruit quality and handling complaints is mostly manual, which can be slow, inconsistent, and sometimes biased. Buyers may receive poor-quality fruits that do not match what was promised, while farmers may face false complaints without proper verification. These gaps highlight the need for an accurate, reliable, and automated solution.

This research focuses on developing an AI-based Fruit Quality Verification and Complaint Validation System that ensures fairness and trust in the smart fruit marketplace. The system will use Convolutional Neural Networks (CNNs) and transfer learning models such as ResNet, MobileNet, or EfficientNet to analyze fruit images uploaded by both farmers before transport and buyers after delivery. To prevent cheating, the system only allows images captured using the in-app camera while also recording metadata such as device ID, GPS location, and timestamp. A complaint handling mechanism is also included, where AI verifies the claim by comparing before- and after-delivery images. If a farmer receives multiple valid complaints, the system will automatically trigger a suspension, improving accountability.

By combining advanced computer vision techniques with backend APIs and database support, the system provides a structured way to track quality, complaints, and user reputation. This research aims to increase trust, transparency, and efficiency in fruit trading, ensuring fair outcomes for both farmers and buyers in the supply chain.

*Keywords – Fruit Quality, Complaint Validation, CNN, Transfer Learning, Smart Supply Chain, AI Verification*

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## LIST OF ABBREVIATIONS

| Abbreviation | Description                                                   |
|--------------|---------------------------------------------------------------|
| AI           | Artificial Intelligence                                       |
| CNN          | Convolutional Neural Network                                  |
| GPS          | Global Positioning System                                     |
| EXIF         | Exchangeable Image File Format (metadata stored in images)    |
| pHash        | Perceptual Hash (used to detect duplicate or tampered images) |
| DB           | Database                                                      |
| API          | Application Programming Interface                             |
| REST         | Representational State Transfer (style of web API)            |
| ID           | Identification / Identifier                                   |
| UI           | User Interface                                                |
| UX           | User Experience                                               |
| QoS          | Quality of Service                                            |
| F1-score     | Harmonic mean of Precision and Recall (AI performance metric) |
| TN           | True Negative (AI correctly identifies non-defective fruit)   |
| TP           | True Positive (AI correctly identifies defective fruit)       |
| VM           | Virtual Machine (used for model training)                     |
| TFLit        | TensorFlow Lite (optimized AI model for mobile devices)       |



# 1. INTRODUCTION

## 1.1 Background Literature

Fruits are an essential part of human diets, and they are also an important source of income for farmers, especially in agricultural countries like Sri Lanka. However, one of the biggest challenges in the fruit supply chain is the problem of quality assurance. Many buyers often complain that the fruits they receive are damaged, spoiled, or not in the expected condition. On the other hand, farmers also argue that they deliver fresh fruits, but damage often happens during transport or due to poor handling after delivery. Because of this, there is a lack of trust between farmers and buyers. This problem affects the reputation of farmers, reduces income, and discourages direct trade in the supply chain.



*Figure 1.1: Fresh fruits packaged by farmers before transport*

In Sri Lanka, post-harvest losses of fruits are reported to be around 30–40%, mostly due to spoilage, mishandling, and poor transport conditions. These issues mean that a large portion of the harvest never reaches the consumer in good condition. When buyers receive low-quality fruits, they often reject them or pay a much lower price. Since there is no standard way to prove the quality at the time of packaging and delivery, both parties tend to blame each other. This lack of objective quality verification has been a long-standing issue in the fruit trade.

Traditional methods of quality checking mainly depend on physical inspections by buyers or intermediaries. These inspections are subjective, slow, and not always accurate. Moreover, in digital trading platforms or store-less distribution systems, it is almost impossible for a buyer to check the fruit quality before delivery. Therefore, there is a strong need for a technological

solution that can provide fair, transparent, and automated fruit quality verification between farmers and buyers.

In recent years, computer vision and artificial intelligence (AI) have been widely used in agriculture for tasks such as fruit detection, ripeness classification, and disease identification. Convolutional Neural Networks (CNNs), a type of deep learning model, have been especially successful in analyzing images and identifying patterns that humans may overlook. For example, MobileNet, ResNet, and EfficientNet are popular CNN architectures used in mobile and low-resource environments because they balance accuracy with speed. These models have been applied in other countries to identify spoiled fruits, check fruit grading standards, and even classify fruits based on ripeness levels. Such advancements show that AI can be an effective tool for fruit quality verification.

Some researchers have proposed systems where farmers upload fruit images before shipment, and those images are processed by AI models to classify the fruits as fresh, spoiled, or damaged. Similarly, buyers can upload images when receiving fruits to verify the actual condition. If there is a mismatch between the two sets of images, the system can help in resolving disputes more fairly. This method reduces subjectivity and builds more trust between parties. However, most of these studies are still at a pilot stage and not widely implemented in developing countries like Sri Lanka.



*Figure 1.2: Damaged fruits received by buyer after transport.*

In addition to AI-based image analysis, another challenge is preventing cheating. If farmers or buyers are allowed to upload any picture, they might upload old or unrelated images. To solve this, modern systems use in-app cameras only, preventing users from selecting pictures from their gallery. Metadata such as GPS location, timestamp, and device ID can also be logged to ensure the authenticity of the image. This way, the system can confirm that the photo was taken at the correct time and location, and it has not been altered.

A further step is to link these AI-verified results with a complaint management system. If a buyer raises a complaint about poor fruit quality, the system can automatically compare the buyer's delivery images with the farmer's packaging images. If the AI confirms the complaint, it will be recorded in the system. Farmers who repeatedly receive valid complaints (for example, more than five times) can be automatically suspended or flagged for review. This creates accountability and ensures that only farmers who maintain good quality are allowed to trade.

Globally, there are already examples of digital agriculture platforms that integrate AI for crop monitoring and quality assurance. However, applying these methods specifically to fruit supply chains in Sri Lanka is still new and innovative. Since Sri Lanka suffers from high fruit wastage and quality disputes, introducing such a system can significantly improve efficiency and fairness in the supply chain. Farmers will benefit from better reputations and more reliable income, while buyers will receive fresher, higher-quality fruits with proof of condition.

In summary, the background literature shows that while fruit quality assurance has always been a challenge, recent advances in AI and secure digital systems provide new opportunities to solve this issue. By using CNN-based image verification, in-app camera restrictions, metadata logging, and automated complaint validation, it is possible to create a fair and transparent system. This is especially important for Sri Lanka, where a large percentage of fruit is wasted and trust between farmers and buyers is often low. The proposed quality assurance module aims to address these problems directly and contribute to a more efficient and trustworthy fruit supply chain.

## **1.2 Literature Survey**

Fruit quality assurance is one of the most critical challenges in agricultural supply chains, especially in countries like Sri Lanka where fruits such as bananas, mangoes, and pineapples are both consumed locally and exported. Many studies have reported that 30–40% of harvested fruits in Sri Lanka are lost before they reach the consumer due to poor handling, transport conditions, and lack of quality monitoring [1]. The absence of proper verification systems creates disputes between farmers and buyers. Farmers usually claim that they deliver fruits in good condition, while buyers often complain that delivered fruits are damaged or spoiled. Since there is no reliable method of verifying the quality at both packaging and delivery stages, these disputes remain unresolved, reducing trust and discouraging direct digital trade.

Traditional quality checks mainly rely on physical inspections by buyers or middlemen. These inspections are highly subjective, time-consuming, and vary depending on the person conducting them. In digital marketplaces where farmers and buyers do not meet face-to-face, such traditional inspection methods are not even possible. Therefore, researchers have turned towards computer vision and machine learning techniques for developing automated fruit quality verification systems.

### **1.2.1. AI-based fruit quality detection**

Machine learning has been widely used in agriculture for fruit classification, ripeness detection, and disease identification. In particular, Convolutional Neural Networks (CNNs) have proven effective in analyzing fruit images due to their ability to automatically learn features such as color, texture, and shape. Naranjo-Torres et al. [2] reviewed deep learning applications in agriculture and highlighted that models like ResNet, MobileNet, and EfficientNet are able to classify fruit defects and diseases with high accuracy. Their findings showed that AI can reduce human subjectivity in quality evaluation, making results faster and more reliable.

Similarly, Li et al. [3] demonstrated the use of CNNs for detecting apple defects such as bruises and rot. Their model achieved higher accuracy than manual inspection, proving that AI systems can detect defects invisible to human eyes. This is especially relevant in supply chains where minor bruises during transport often lead to spoilage, and farmers or buyers may deny responsibility. By applying CNNs, quality can be assessed objectively, helping in

fair dispute resolution.

In Sri Lanka, studies have also emphasized the importance of improving fruit grading and quality control. Jayasundara et al. [4] discussed traceability systems for Sri Lankan fruit exports, pointing out that lack of transparent quality verification reduces competitiveness in global markets. Their research highlighted the use of GPS and blockchain for ensuring that exported fruits are tracked from farm to shipment. Such systems, combined with AI-based defect detection, could greatly improve accountability within the supply chain.

### **1.2.2. Image processing and traditional ML approaches**

Apart from deep learning, researchers have also tested classical image processing and machine learning algorithms for fruit quality analysis. Features such as color histograms, texture descriptors like Gray Level Co-occurrence Matrix (GLCM), and shape features have been used in combination with classifiers such as Support Vector Machines (SVM), Random Forests (RF), and k-Nearest Neighbors (k-NN).

For example, Patel et al. [5] applied GLCM with SVM to classify mango ripeness levels and achieved over 80% accuracy. Although not as powerful as CNNs, such models are computationally lighter and easier to deploy on mobile devices. Another study by Silva et al. [6] showed that Random Forest models could be used to detect pineapple spoilage based on texture patterns. These approaches indicate that combining traditional ML with deep learning could produce balanced systems for low-resource environments like rural Sri Lanka.

### **1.2.3. Complaint management and fraud prevention**

A unique aspect of quality assurance in digital supply chains is the handling of complaints and prevention of fraud. In e-commerce, buyers can raise disputes when goods are not as described. Perera et al. [7] proposed a complaint validation framework that used metadata such as timestamps, device IDs, and system logs to verify whether a complaint was genuine. This concept can be directly applied to agriculture: if a buyer claims poor fruit quality, the system can automatically cross-check photos and metadata to determine the validity of the complaint.

In addition, in-app camera enforcement has been widely studied as a method to prevent cheating. Singh et al. [8] showed that restricting users to capture images only from the in-app camera significantly reduced fraudulent uploads in online proctoring systems. Applying the same idea to fruit supply chains prevents farmers or buyers from uploading old or unrelated photos. Metadata such as GPS and timestamps further confirm that the photo was taken at the correct location and time.

To prevent image manipulation, perceptual hashing (pHash) has been used in many domains. Zauner [9] explained that pHash can reliably detect duplicate or altered images by comparing visual similarity instead of exact pixels. In the context of fruit quality verification, pHash could be used to identify whether the same image has been reused across multiple complaints or submissions, thereby reducing fraudulent behavior.

#### **1.2.4. Towards integrated systems**

Although AI-based fruit defect detection, metadata verification, and complaint management have been studied separately, very few systems combine all these elements into a single platform. Most research focuses only on classification accuracy or grading, without addressing real-world challenges such as disputes, fraud, and accountability. For Sri Lanka, where farmers and buyers often struggle with mistrust and post-harvest losses, an integrated system is highly valuable.

Wimalaratana et al. [1] stressed that post-harvest losses in Sri Lanka are not only an economic issue but also a social problem, as they reduce farmer income and limit food security. By combining AI verification with complaint validation and reputation tracking, the FreshRoute system can directly address these challenges. Farmers will have incentives to maintain quality, while buyers gain confidence that their complaints will be fairly handled.



### 1.2.5. Summary of related works

| Research Paper                                                                   | Technique Used                            | Focus                                | Primary Insight                                                 |
|----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------|-----------------------------------------------------------------|
| Reducing post-harvest fruit loss in Sri Lanka's supply chains                    | Survey of fruit supply chain in Sri Lanka | Post-harvest loss and quality issues | 30–40% fruit loss due to poor handling and lack of verification |
| A review of deep learning in agriculture: Techniques and applications            | CNNs (ResNet, MobileNet, EfficientNet)    | Fruit disease and defect detection   | CNNs effective in agricultural image classification             |
| Apple defect detection using deep convolutional networks                         | CNN for Apple Defect Detection            | Detecting bruises and rot in apples  | Achieved high accuracy, outperforming manual inspection         |
| Traceability systems for Sri Lanka's fruit exports: Challenges and opportunities | GPS + Blockchain Traceability             | Tracking Sri Lankan fruit exports    | Improved accountability through metadata and blockchain         |
| Mango ripeness classification using GLCM and SVM                                 | GLCM + SVM                                | Mango ripeness classification        | Over 80% accuracy, lightweight alternative to CNNs              |
| Texture-based pineapple spoilage detection using Random Forests                  | Random Forest (RF)                        | Pineapple spoilage detection         | Texture-based ML for quality detection                          |
| Complaint validation using metadata in e-commerce platforms                      | Metadata in complaint validation          | E-commerce disputes                  | Fraud reduced using timestamps and device IDs                   |

|                                                                    |                            |                                     |                                                    |
|--------------------------------------------------------------------|----------------------------|-------------------------------------|----------------------------------------------------|
| Securing mobile applications using in-app camera enforcement       | In-app camera enforcement  | Preventing cheating in digital apps | Blocked gallery uploads, ensured real-time capture |
| Implementation and benchmarking of perceptual image hash functions | Perceptual Hashing (pHash) | Image forgery detection             | Reliable in spotting duplicate or tampered images  |

*Table 1.1: Overview of existing methods and technologies*

### 1.3 Research Gap

The reviewed studies on fruit quality verification and supply chain monitoring show that researchers have made good progress in applying machine learning (ML) and image processing to detect fruit defects, classify ripeness levels, and reduce human subjectivity in quality checks. CNNs, SVMs, and image processing methods like GLCM have already demonstrated high accuracy in classifying fruit conditions [2] [3] [5]. In addition, traceability frameworks using GPS and blockchain have been tested to improve accountability in agricultural exports from Sri Lanka [4]. Complaint validation methods from e-commerce also show promise for adapting to agriculture [7].

However, several important research gaps remain, especially for real-world applications in Sri Lanka's fruit supply chain.

One major gap is that most fruit quality detection systems only focus on classification accuracy but do not directly handle disputes between farmers and buyers. Even if AI can detect spoiled or fresh fruit, these systems are not connected to a proper complaint validation framework. In Sri Lanka, where trust between farmers and buyers is often low due to repeated disputes, an integrated complaint-handling system is needed [1].

Another gap is that many existing systems allow image uploads from the gallery, making them vulnerable to fraud. Farmers or buyers can upload old images instead of real-time captures, which reduces reliability. Research in other fields like online exams has shown that enforcing in-app camera use prevents cheating [8], but this approach has not yet been applied to agriculture in Sri Lanka.

Metadata logging is also underused. While studies highlight the importance of traceability in exports [4], most fruit quality systems do not include metadata such as GPS, timestamps, and device IDs when capturing images. This limits their effectiveness in proving the authenticity of complaints. A proper system should ensure that every photo has metadata attached so it can be verified later in disputes.

Another key issue is that very few studies combine AI-based fruit verification with reputation

tracking and automatic suspension mechanisms. In e-commerce and ride-hailing platforms, multiple valid complaints can reduce a seller's rating or even suspend them [7]. However, in agriculture, such automated reputation systems are not widely researched. This is a missed opportunity, because accountability through reputation scoring could encourage both farmers and buyers to behave fairly.

Finally, while Sri Lanka has research on post-harvest losses and supply chain inefficiencies [1] [6], there is a lack of applied AI solutions tailored to local challenges. Most systems are either imported or tested in developed countries, meaning they may not work well in local rural contexts where internet and smartphone usage differ.

In summary, the current literature provides valuable tools such as CNN-based classification, metadata logging, and complaint validation separately, but does not integrate them into a single solution. The proposed system addresses this by combining AI-based fruit image comparison, in-app camera enforcement, metadata logging, pHash-based fraud prevention, complaint validation, and automated suspension into one comprehensive module for Sri Lanka's fruit supply chain. This integration will fill the research gap and provide a more practical and trustworthy solution for farmers, buyers, and digital marketplaces

| <b>Feature / Gap</b>                         | [1] | [2] | [3] | [4]     | [5] | <b>Proposed Solution</b> |
|----------------------------------------------|-----|-----|-----|---------|-----|--------------------------|
| AI-based fruit defect detection              | Yes | Yes | Yes | No      | Yes | Yes                      |
| Metadata logging (GPS, device ID, timestamp) | No  | No  | No  | Yes     | No  | Yes                      |
| In-app camera enforcement                    | No  | No  | No  | No      | No  | Yes                      |
| Complaint validation system                  | No  | No  | No  | No      | Yes | Yes                      |
| Reputation tracking & auto-suspension        | No  | No  | No  | No      | No  | Yes                      |
| Integrated platform for Sri Lanka            | No  | No  | No  | Partial | No  | Yes                      |

*Table 1.2: Comparison of the features addressed by the previous research studies*

## **2. RESEARCH PROBLEM**

Fruit is one of the most important agricultural products in Sri Lanka, both for local consumption and for export. However, poor handling, lack of monitoring, and absence of proper quality assurance cause a large percentage of fruits to be damaged or spoiled before reaching consumers. Studies show that post-harvest fruit loss in Sri Lanka can be as high as 30–40% [1]. This not only reduces farmer income but also increases disputes between farmers and buyers. Farmers often claim that they send good-quality fruits, while buyers complain that delivered fruits are damaged. Since there is no reliable way to verify these claims, trust within the supply chain is reduced.

The current process of fruit quality checking is mostly manual, based on visual inspection. Buyers or middlemen physically check fruits before accepting or rejecting them. This process is subjective, time-consuming, and often influenced by personal judgment. In digital marketplaces, where farmers and buyers do not meet face-to-face, such inspections are not even possible. Therefore, many disputes remain unresolved, discouraging direct digital trade and weakening the adoption of smart agriculture platforms [2].

Another problem is the possibility of cheating or fraud when farmers or buyers upload images. In most existing systems, users can upload photos from their gallery, which means old or unrelated images can be used to misrepresent fruit quality. This creates unfair advantages and reduces the credibility of complaint handling mechanisms. Research has shown that enforcing in-app camera use and metadata logging (time, location, device ID) can reduce this problem [8], but these methods are not yet applied to Sri Lankan fruit supply chains.

Currently, there is no integrated system that combines AI-based fruit quality verification with complaint validation and reputation tracking. Some studies focus only on classification accuracy of fruits [2] [3], while others look at traceability for exports [4]. However, they do not address the full cycle of verifying, validating complaints, and updating farmer or buyer reputation. This gap leaves room for fraud, disputes, and lack of accountability.

Therefore, there is a clear research problem: how to design and implement an AI-based fruit quality verification and complaint validation system that is reliable, tamper-proof, and suitable for the Sri Lankan fruit supply chain. This problem must be addressed to improve trust, reduce disputes, and encourage wider adoption of smart digital marketplace.

### **3. RESEARCH OBJECTIVES**

#### **3.1 Main Objective**

The main objective of this research is to design and implement an AI-based fruit quality verification and complaint validation system that can increase trust and accountability in Sri Lanka's fruit supply chain. The proposed system will focus on preventing cheating during image uploads by enforcing in-app camera use, automatically collecting metadata (GPS, timestamp, device ID), and applying image comparison techniques using transfer learning models such as ResNet, MobileNet, or EfficientNet. The system will also include a complaint-handling process, where buyers can submit complaints with images, and the AI will validate whether the complaint is genuine by comparing pre-transport and post-delivery fruit images.

If the complaint is valid, the system will update the farmer's reputation score, and after several valid complaints, the farmer's account will be temporarily suspended. By doing so, the solution will encourage both farmers and buyers to behave fairly, while reducing disputes and improving the efficiency of digital agriculture marketplaces. Ultimately, the project aims to provide a reliable and transparent quality assurance component that can be integrated into larger smart agriculture platforms in Sri Lanka.



## **3.2 Sub Objectives**

### **Sub Objective 1: To Develop an Accurate AI Model for Fruit Quality Verification**

This sub-objective focuses on building and training a machine learning model that can correctly identify differences between fresh fruit images (uploaded by farmers before transport) and delivered fruit images (uploaded by buyers after transport). Transfer learning methods such as ResNet, MobileNet, or EfficientNet will be applied to achieve better accuracy with limited training data. The AI model will be able to classify whether fruits are in good condition, spoiled, or damaged. This step is important because it replaces manual inspections that are often slow, subjective, and impractical in online marketplaces. A reliable AI model will ensure that both farmers and buyers have a fair and consistent quality-checking process.

### **Sub Objective 2: To Enforce Secure Image Capture Using In-App Camera**

One of the major problems in complaint systems is cheating, where users upload old or unrelated images from their gallery. To prevent this, the system will enforce image capture only through the in-app camera. By blocking gallery uploads, every image will be taken in real-time at the moment of packaging or delivery. This reduces fraud and increases trust in the platform. Similar approaches have been used in online exam systems to prevent cheating, and adapting this method for agriculture is expected to improve fairness in complaint handling.

### **Sub Objective 3: To Implement Metadata Logging for Authenticity**

This sub-objective aims to capture and store important metadata such as GPS location, timestamp, and device ID every time an image is uploaded. Metadata will be stored in the database along with the image so that disputes can be resolved more transparently. For example, if a buyer complains that the fruit was damaged, the system can verify when and where the image was taken. Metadata adds a strong layer of accountability, ensuring that neither party can easily manipulate or fake the process. This will be particularly useful in Sri Lanka, where disputes often occur during transport due to delays or poor handling.

#### **Sub Objective 4: To Design a Complaint Validation and Reputation Tracking Mechanism**

This sub-objective focuses on designing a complaint-handling system that automatically validates buyer complaints using AI image comparison. If the system confirms that the delivered fruit is of lower quality than the pre-transport image, the complaint will be recorded as valid. Over time, the system will track valid complaints against farmers or buyers and maintain a reputation score. After five valid complaints, the farmer's account will be temporarily suspended to protect buyers and encourage accountability. This system creates a fair environment by discouraging fraud and rewarding genuine transactions. It also builds long-term trust between farmers and buyers in digital marketplaces.

#### **Sub Objective 5: To Integrate the System with Existing Marketplace Platforms**

For real-world use, the system must not work in isolation but instead connect with existing e-commerce or agriculture marketplace platforms. This sub-objective aims to create backend APIs and a database that can easily be integrated with other modules, such as forecasting, logistics, and marketplace payment systems. Integration ensures that quality verification and complaint handling become part of the normal transaction workflow, making the system practical and scalable. This step also aligns with Sri Lanka's growing trend of digital agriculture platforms that aim to connect rural farmers with urban buyers.

#### **Sub Objective 6: To Evaluate and Improve System Accuracy and Usability**

Finally, the system must be tested and evaluated under real conditions. This sub-objective focuses on testing the AI model with fruit images collected from local farmers and buyers in Sri Lanka. Performance will be measured using accuracy, precision, recall, and F1-score to ensure the AI can make fair decisions. Usability testing will also be carried out to see if farmers and buyers can use the mobile app easily. Based on feedback, improvements will be made to the interface and model performance. The goal is to create a system that is not only technically accurate but also user-friendly, reliable, and trusted by the agricultural community.

## **4. METHODOLOGY**

The development of the proposed AI-based Fruit Quality Verification and Complaint Validation System will follow a step-by-step process to ensure accuracy, reliability, and user-friendliness. The methodology outlines how the system will be designed, implemented, and tested to meet the needs of farmers, buyers, and marketplace administrators in Sri Lanka's fruit supply chain. The main phases of the methodology are as follows.

### **4.1 Data Collection**

The first step is to collect a suitable dataset of fruit images. Farmers will capture images of fruits before transport using the in-app camera, while buyers will capture images after delivery. To train the AI model, additional fruit datasets will be collected from open-source repositories and local markets in Sri Lanka. The dataset will include different fruit types such as mangoes, bananas, papayas, and pineapples, covering both fresh and damaged conditions. Collecting real-world images is important to represent variations in lighting, angles, and fruit handling, which are common in practical scenarios.

### **4.2 Data Pre-processing**

Once collected, the images will be pre-processed to make them consistent and ready for training. Pre-processing will include resizing images, normalizing pixel values, and applying data augmentation techniques such as rotation, flipping, and brightness adjustments. These steps will make the dataset more diverse and prevent the model from overfitting. For example, images taken in low light or at odd angles should still be recognizable by the system. Pre-processing ensures that the AI model can perform well in real-world conditions, where image quality may not always be perfect.

### **4.3 AI Model Development**

The next stage is to develop a machine learning model capable of comparing fruit images before and after delivery. Transfer learning methods such as ResNet, MobileNet, or EfficientNet will be used to save time and improve accuracy, since these models are already trained on large image datasets. The model will be fine-tuned with our pre-processed fruit

dataset to detect changes in freshness, texture, and visible damage.

The model will be trained to output whether the fruit is “Good Quality,” “Damaged,” or “Spoiled.” By comparing the farmer’s image (before transport) with the buyer’s image (after delivery), the model can identify whether a valid complaint exists. Training will involve iterative testing, tuning hyperparameters, and evaluating performance using metrics such as accuracy, precision, recall, and F1-score.

#### **4.4 In-App Camera Integration**

To prevent cheating, the system will strictly enforce image capture through the in-app camera. This means users will not be able to upload images from their gallery. Instead, they must take real-time photos using the app. This step reduces the chances of fraud, where users might otherwise submit old or fake images. This feature will be implemented during the mobile app development phase, ensuring that both farmers and buyers follow the same verification process.

#### **4.5 Metadata Logging**

Every image captured through the in-app camera will automatically store important metadata such as GPS location, timestamp, and device ID. Metadata will be saved in the database along with the image. This ensures that the system can verify exactly when and where the photo was taken, adding an additional layer of security and transparency. For example, if a buyer claims that fruits were damaged during transport, metadata can confirm whether the photo was truly taken at the delivery point.

#### **4.6 Complaint Validation Mechanism**

The heart of the system is the complaint validation module. When a buyer files a complaint, the system will compare the pre-transport image (from the farmer) with the post-delivery image (from the buyer). The AI model will analyze differences in fruit condition and decide whether the complaint is valid.

If the complaint is found valid, it will be recorded in the system, and the farmer’s reputation score will be updated. If a farmer receives five valid complaints, their account will be

automatically suspended for investigation. This discourages poor handling practices and encourages accountability. On the other hand, if the complaint is invalid, the buyer's account may also be flagged to prevent false reporting.

#### **4.7 Backend APIs and Database Development**

To manage all images, complaints, and metadata, a secure backend system will be developed. RESTful APIs will connect the mobile app with the database, ensuring smooth communication. The database will store user accounts, fruit images, complaint records, metadata logs, and reputation scores. APIs will also allow the quality assurance system to integrate with other modules of the digital marketplace, such as order management and forecasting systems.

#### **4.8 User Interface Design**

A simple and user-friendly mobile interface will be created for both farmers and buyers. The app will guide users step by step to capture images, file complaints, and view reputation scores. The design will prioritize usability, ensuring that farmers with low technical knowledge can still use the system easily. For administrators, a dashboard will provide an overview of valid complaints, reputation tracking, and flagged accounts.

#### **4.9 Testing and Validation**

After development, the system will undergo extensive testing to ensure it performs well in real-world conditions. Testing will include:

- ❖ AI model testing with local fruit images to measure accuracy.
- ❖ Functional testing to verify metadata logging, in-app camera restrictions, and complaint validation.
- ❖ Usability testing with farmers and buyers to ensure the app is easy to use.
- ❖ Feedback from initial testing will be used to improve the model and the app's design.

#### **4.10 Prototype Launch and Evaluation**

Finally, a prototype version of the system will be launched and evaluated in collaboration with local farmers and buyers. Real transactions will be monitored to assess how well the

system prevents disputes and builds trust. The prototype results will guide further improvements before scaling up for wider adoption.

#### 4.11 System Architecture Diagram

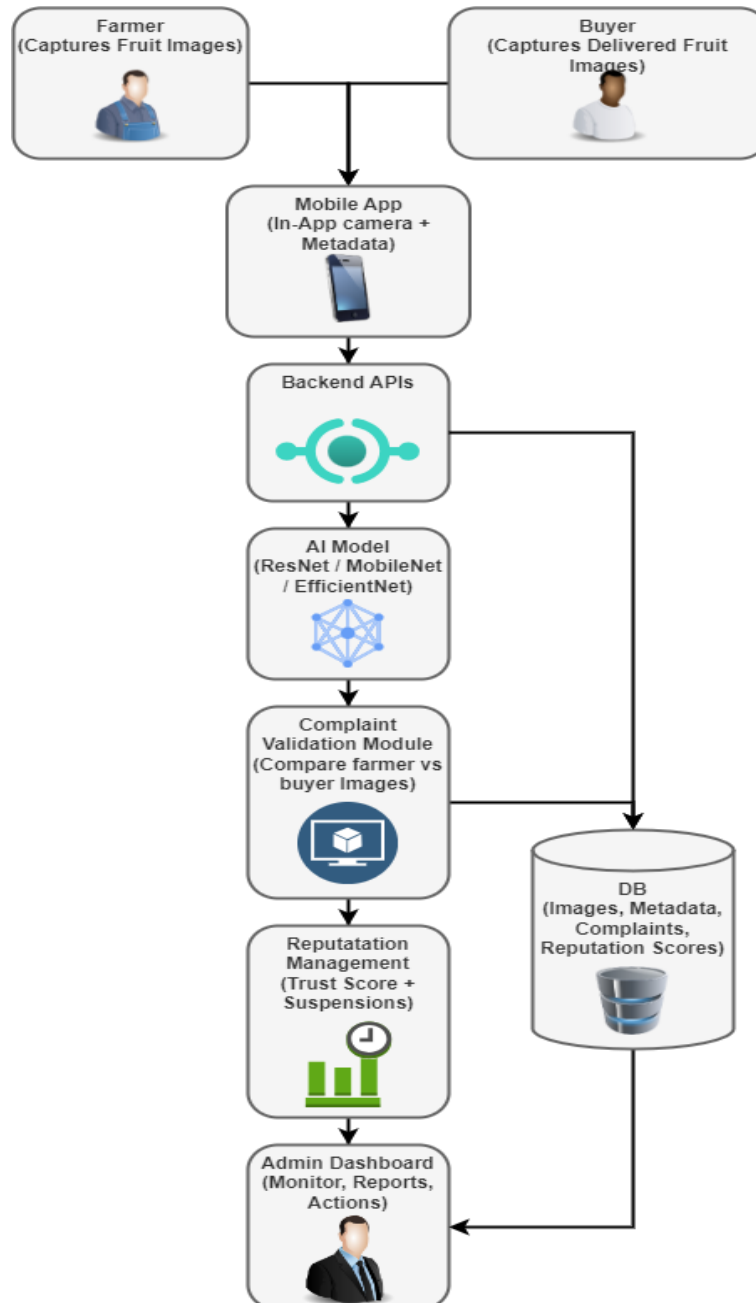


Figure 4.1: Individual component system architecture diagram

The system architecture for the Fruit Quality Verification and Complaint Validation System is designed to ensure that fruit images captured before transport and after delivery are fairly analyzed using AI. The workflow starts with the farmer, who must capture images of the fruit before dispatch using the in-app camera. These images are sent to the backend server where they are stored together with metadata such as timestamp, GPS location, and device ID.

When the buyer receives the fruit, they also capture images through the same in-app camera. These new images are then sent to the system and matched with the corresponding farmer-uploaded images. The AI model, trained with transfer learning (ResNet/MobileNet/EfficientNet), compares both sets of images to detect differences such as damage, spoilage, or reduced quality.

The results of this analysis are processed by the complaint validation module. If the AI finds significant differences proving that the fruits are damaged, the complaint is marked as valid. This decision, along with the supporting image evidence and metadata, is stored in the database for record keeping. Farmers who receive repeated valid complaints (five or more) will be automatically flagged or suspended, while buyers making false complaints may also be restricted.

A reputation management system is integrated within the architecture to track the trust scores of both farmers and buyers. This makes it easy for the marketplace to identify reliable sellers and honest buyers.

The backend APIs enable communication between the mobile application and the server. The database stores user profiles, complaint history, metadata logs, and AI results. On the user side, the mobile interface allows farmers and buyers to easily upload images, check complaint status, and view their reputation score. An admin dashboard provides the system administrator with tools to monitor complaints, generate reports, and manage flagged accounts.

In the end, this architecture ensures fairness, prevents cheating (since only in-app photos are allowed), and increases trust between farmers and buyers in the fruit supply chain.

## 5. PROJECT REQUIREMENTS

### 5.1 Functional Requirements

- ❖ **In-app Image Capture** - The application must allow both farmers and buyers to capture fruit images only through the in-app camera. This ensures that no pre-stored or edited images from the gallery are uploaded, reducing the chances of cheating.
- ❖ **Metadata Logging** - Every image captured must include metadata such as device ID, timestamp, and GPS location. This will help in verifying that the image is authentic and taken at the correct time and place.
- ❖ **AI-Based Image Comparison** - The system must use AI algorithms (such as ResNet, MobileNet, or EfficientNet) to compare fruit images uploaded by farmers before delivery and by buyers after receiving the goods. The AI should detect differences that may indicate damage, poor quality, or fraud.
- ❖ **Complaint Registration** - The application must allow buyers to submit a complaint if they believe the quality of delivered fruits does not match what was promised. Farmers must also be notified of the complaint immediately.
- ❖ **Complaint Validation System** - An AI-driven module must validate the complaint by analyzing and comparing the two sets of images (before and after delivery). Only valid complaints should be approved to avoid false claims.
- ❖ **Reputation and Suspension Management** - The system must automatically update farmer reputation scores based on the number of valid complaints received. If a farmer receives five valid complaints, the system must trigger an automatic suspension of their selling privileges.
- ❖ **Database Storage** - The application must securely store fruit images, complaint records, farmer/buyer profiles, and reputation history in a central database for tracking and reporting.
- ❖ **Admin Dashboard** - There must be an admin dashboard for monitoring complaints, reputation scores, farmer suspensions, and generating reports for future analysis.
- ❖ **System Integration** - The complaint verification system must integrate seamlessly with the main Smart Fruit Supply Chain platform, including the marketplace and forecasting modules.



## 5.2 Non-Functional Requirements

### ❖ Reliability

The system must provide consistent and accurate complaint validation results. Wrong decisions (such as rejecting a genuine complaint or approving a false one) could damage trust in the platform. Therefore, the AI models must be thoroughly trained and tested to ensure reliable performance.

### ❖ Security

Since the system involves sensitive data such as user identities, device IDs, GPS locations, and complaint records, security must be a top priority. Data encryption, secure authentication, and controlled access must be implemented. Farmers and buyers should only access their own data, and admins should have higher privileges.

### ❖ Availability

The system must be available 24/7 since farmers and buyers may operate at different times. Any downtime could cause delays in complaint submissions and validations, reducing trust in the system. The backend infrastructure should be designed for high availability with backup mechanisms.

### ❖ Usability

The application must be simple and easy to use for both farmers and buyers. Many farmers may have limited technical skills, so the system should include clear instructions, simple navigation, and visual guidance for capturing fruit images. The admin dashboard should also be user-friendly with clear reports and complaint tracking.

### ❖ Scalability

As the platform grows, the number of farmers, buyers, and complaint cases will increase. The system must be able to handle a growing number of users, image uploads, and AI processing without reducing performance. The backend should support horizontal scaling (adding more servers) when demand increases.

#### ❖ **Performance**

Image comparison and complaint validation must happen quickly to avoid delays. Long processing times could frustrate users. The AI models should be optimized (e.g., using TensorFlow Lite for mobile devices) to provide fast responses even when running on smartphones.

#### ❖ **Maintainability**

The system should be built in a modular way so that AI models, APIs, and databases can be updated without affecting the entire platform. This makes it easier to fix bugs, add new features, or improve AI algorithms in the future.

#### ❖ **Transparency and Fairness**

Users (farmers and buyers) must trust the system. Therefore, the complaint validation results should be explainable in simple terms. For example, the system should provide a short explanation like “Complaint valid: Fruit showed visible damage compared to pre-transport images.” This builds confidence among users.

### **5.3 Personal Requirements**

#### ❖ **Guidance from supervisors on the fruit supply chain domains**

This component of the project requires interdisciplinary guidance, combining expertise from agriculture (fruit quality standards), supply chain management, and advanced AI/data science techniques. Supervisors with agricultural knowledge can provide insights on fruit grading, common quality issues, and practical standards followed in Sri Lanka. Similarly, technical supervisors with AI and machine learning experience will help in selecting the right models (such as ResNet, MobileNet, or EfficientNet) for fruit image classification and comparison. Their guidance is also required in designing the complaint validation system, ensuring that the solution is fair, accurate, and implementable in real-world conditions. This combination of agricultural and technical expertise will ensure the system is both scientifically sound and practically relevant.

❖ **Understanding the specific area of each domain to proceed with the selected research area**

To successfully implement this component, it is important to build a strong understanding of fruit quality verification methods, complaint handling mechanisms, and AI-based image processing. From the agriculture side, knowledge of fruit defects, storage conditions, and transportation issues is necessary to define the parameters that the AI model should detect. From the AI side, familiarity with image preprocessing, transfer learning, and performance evaluation techniques is required to train a model that is robust and reliable. Additionally, an understanding of supply chain practices in Sri Lanka, including farmer–buyer interactions, complaint mechanisms, and trust management, is crucial for integrating the solution into the larger Smart Fruit Supply Chain platform. This interdisciplinary awareness will help in developing a solution that not only works technically but also aligns with real-world farmer and buyer needs.

## 5.4 Expected Test Cases

In the development and implementation of the Fruit Quality Verification and Complaint Validation System, several test cases will be carried out to check accuracy, usability, and overall performance. These test cases ensure that the system works properly for farmers, buyers, and platform administrators. The following are the key expected test cases.

### i. Fruit image uploading and preprocessing test

|                         |                                                                                                                                                                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>        | To verify that the system correctly accepts, processes, and prepares fruit images for analysis.                                                                                                                                                         |
| <b>Test Case</b>        | Upload fruit images in different formats (JPEG, PNG) and varying resolutions directly using the in-app camera. Check if the system rejects gallery images to prevent cheating and correctly preprocesses the photos (resizing, normalizing, enhancing). |
| <b>Expected Outcome</b> | The system should only accept camera-captured images, preprocess them successfully, and standardize all inputs without errors.                                                                                                                          |

*Table 5.1: Test case 1*

### ii. Fruit quality verification test

|                         |                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>        | To validate the accuracy of the model in detecting fruit quality (good vs. damaged).                                                   |
| <b>Test Case</b>        | Use a dataset of test images with both fresh and damaged fruits. Compare the system's classification output with human expert ratings. |
| <b>Expected Outcome</b> | The system should classify fruit images accurately (fresh/defected) with results matching the ground truth labels.                     |

*Table 5.2: Test case 2*

iii. Complaint validation test

|                         |                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>        | To check if the system can validate buyer complaints by comparing farmer-uploaded and buyer-uploaded fruit images.                                         |
| <b>Test Case</b>        | Submit complaints with before-delivery and after-delivery fruit images. Verify that the AI compares both sets and confirms whether the complaint is valid. |
| <b>Expected Outcome</b> | The system should accurately validate complaints, approving genuine ones and rejecting false claims.                                                       |

*Table 5.3: Test case 3*

iv. Complaint history and suspension rule test

|                         |                                                                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>        | To ensure that the system keeps proper complaint history and applies automatic suspension after repeated valid complaints.                                               |
| <b>Test Case</b>        | Submit multiple complaints against a farmer account. Verify that the system updates complaint records in the database and suspends the account after 5 valid complaints. |
| <b>Expected Outcome</b> | The system should track all complaints accurately and automatically suspend accounts after the threshold is reached.                                                     |

*Table 5.4: Test case 4*

v. Metadata logging test

|                         |                                                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>        | To check if the system logs all metadata correctly (GPS location, timestamp, device ID).                                          |
| <b>Test Case</b>        | Capture images from different devices at different times and locations. Verify that metadata is stored properly with each upload. |
| <b>Expected Outcome</b> | The system should record accurate metadata for every image, ensuring transparency and preventing fraud.                           |

*Table 5.5: Test case 5*

vi. Marketplace integration test

|                         |                                                                                                                                                                                               |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>        | To validate the smooth integration of fruit quality verification and complaint validation modules with the Smart Fruit Supply Chain marketplace.                                              |
| <b>Test Case</b>        | Upload fruit images, verify quality, and process complaints through the marketplace interface. Check if data flows correctly between quality assurance, forecasting, and marketplace modules. |
| <b>Expected Outcome</b> | The system should fully integrate with the platform, sharing complaint data, fruit quality status, and user reputation seamlessly.                                                            |

*Table 5.6: Test case 6*

vii. Performance and scalability test

|                         |                                                                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>        | To evaluate how the system handles multiple users, large image uploads, and increasing complaint requests.                                                              |
| <b>Test Case</b>        | Simulate a large number of farmers and buyers uploading images and submitting complaints at the same time. Monitor system performance, response times, and server load. |
| <b>Expected Outcome</b> | The system should remain stable, process requests quickly, and scale effectively under high demand without crashing.                                                    |

*Table 5.7: Test case 7*

## **6. COMMERCIALIZATION**

The launch of the Fruit Quality Verification and Complaint Validation System can bring a major improvement to the way fresh fruits are traded between farmers and buyers. This solution makes use of artificial intelligence to check fruit quality and validate complaints in a fair and transparent way. By doing so, it reduces fraud, builds trust between farmers and buyers, and ensures that only genuine complaints are accepted. The system is designed to directly integrate with the Smart Fruit Supply Chain platform, which can be adopted by local farmers, fruit traders, and export companies.

For commercialization, the system can be offered through a subscription-based model, where farmers and buyers pay a small monthly or annual fee to use the service within the marketplace. Export companies and large-scale buyers can be offered enterprise packages that provide advanced features such as detailed quality reports, complaint analytics, and reputation scoring. This will ensure a sustainable revenue model while also keeping the system affordable for small farmers.

Additionally, the system can be extended as a mobile and web application that connects directly with digital marketplaces. By doing so, it can be scaled across Sri Lanka and later expanded to other agricultural markets in South Asia. The AI-based verification not only improves fairness in transactions but also supports farmers in maintaining a positive reputation, helping them secure long-term business partnerships.

To promote adoption, the system can be marketed through agricultural exhibitions, government farmer support programs, and cooperative societies. Partnerships with agricultural boards and export authorities can also help in expanding reach. By highlighting benefits such as fraud prevention, improved trust, and fair trade, the system has the potential to become a standard tool in modern fruit supply chain management.

## 7. SOFTWARE SPECIFICATIONS

The development of the Fruit Quality Verification and Complaint Validation System will use a set of modern tools and technologies to make sure the platform is reliable, easy to use, and scalable for farmers and buyers. The selected software stack supports both machine learning for fruit quality checking and web/mobile applications for users.

|                         | <b>Tools and Technologies</b>                 | <b>Purpose</b>                                                    |
|-------------------------|-----------------------------------------------|-------------------------------------------------------------------|
| Programming languages   | Python                                        | for machine learning models and backend logic                     |
|                         | JavaScript                                    | for frontend and interactive features.                            |
|                         | Tailwind / MUI                                | for structuring and styling the web application                   |
| Frameworks              | TensorFlow / PyTorch                          | training and deploying the fruit quality verification model.      |
|                         | FastAPI / Flask                               | building APIs and backend services.                               |
|                         | React.js / ReactNative                        | Frontend development.                                             |
| Libraries               | OpenCV                                        | image processing for fruit defect detection.                      |
|                         | Pandas / Numpy                                | data analysis and preprocessing.                                  |
| Database                | Firebase / MongoDB                            | store user profiles, complaints, and validation reports securely. |
| APIs                    | REST API                                      | communication between the app, backend, and database.             |
| Development environment | Visual Studio Code / PyCharm / Android Studio | Primary IDEs                                                      |
|                         | GitHub                                        | Version controlling                                               |

*Table 7.1: Tools and technologies*

These tools make the system efficient and secure, ensuring smooth image processing, complaint validation, and storage of important transaction data.



## 8. BUDGET

| Description                                  | Amount (LKR) |
|----------------------------------------------|--------------|
| Web Hosting & Domain                         | 3,500.00     |
| Model Training Resources (GPU/Colab Pro)     | 12,000.00    |
| SMS / Notification Gateway Credits           | 5,000.00     |
| Cloud Services                               | 5,500.00     |
| Miscellaneous (transport, printing, testing) | 10,000.00    |
| Total Estimated Budget                       | 36,000.00    |

*Table 8: Budget table*

This budget covers the essential costs for developing the Fruit Quality Verification and Complaint Validation System. The largest allocation is for Model Training Resources (LKR 12,000) to ensure the AI models are properly trained for fruit quality verification and complaint validation.

Web Hosting & Domain (LKR 3,500) and Cloud Services (LKR 5,500) are needed to deploy and maintain the system online so that users can access it reliably. SMS / Notification Gateway Credits (LKR 5,000) are included to allow automated alerts and complaint updates to reach farmers and buyers, even in rural areas where internet access may be limited.

Finally, a Miscellaneous fund (LKR 10,000) is reserved for unexpected expenses such as transport for field testing, printing, and other project needs.

This streamlined budget ensures all critical requirements are covered while keeping the project cost-effective and practical.

## 9. WORK BREAKDOWN STRUCTURE

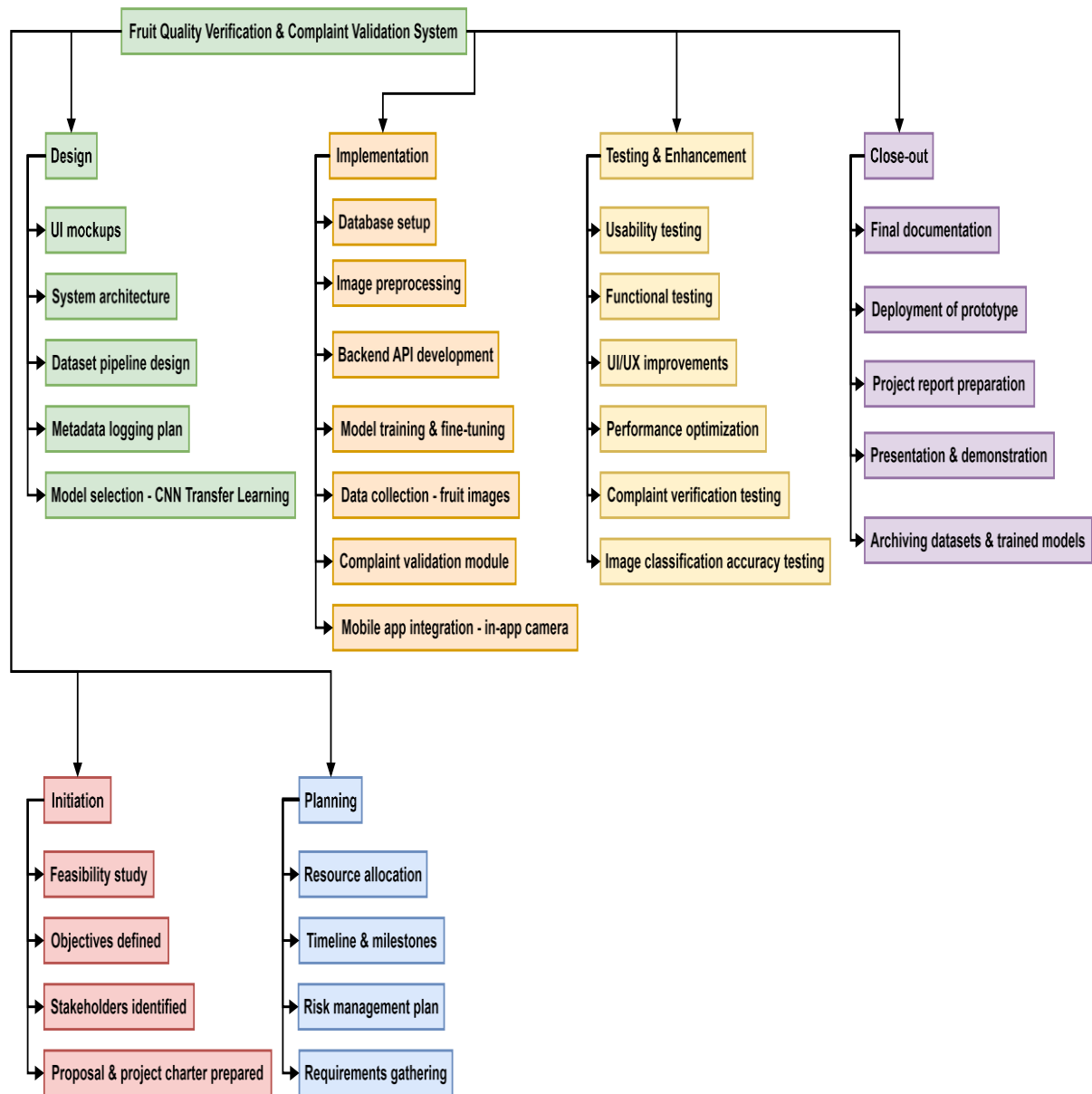


Figure 9.1: Work breakdown structure

The work breakdown structure (WBS) for the Fruit Recognition and Classification System is arranged as a clear hierarchy, showing all the important phases and sub-tasks required to complete the project successfully. It is divided into six main phases: initiation, planning, design, implementation, testing & enhancement, and close-out.

In the **initiation phase**, the basic groundwork of the project is carried out. This includes

identifying the main objectives, understanding the problem domain, and preparing the initial proposal and project charter. Stakeholders are recognized, and feasibility of the project is also reviewed here.

The **planning phase** defines the exact requirements and allocates resources. It covers setting timelines, assigning responsibilities, and preparing a risk management plan. This ensures the project has a clear direction before moving forward.

The **design phase** focuses on creating the architecture of the system. Tasks in this phase include designing the dataset pipeline, selecting machine learning models, planning preprocessing steps, and creating mockups for the user interface.

The **implementation phase** involves the actual building of the system. Sub-tasks include collecting and cleaning fruit images, preprocessing data, training the model, developing the backend API, and integrating the frontend interface.

In the **testing & enhancement phase**, the system is validated for accuracy, reliability, and usability. Test cases are executed to ensure the system meets functional requirements. Enhancements are added, such as improving model accuracy, optimizing system performance, and refining the UI for better user experience.

Finally, the **close-out phase** includes preparing the final documentation, compiling project reports, and presenting the outcomes. It also involves storing datasets, trained models, and source code for future use.

By structuring the work into these phases and sub-tasks, the project can be managed in a step-by-step manner, ensuring all goals are completed smoothly.

## 10. GANTT CHART

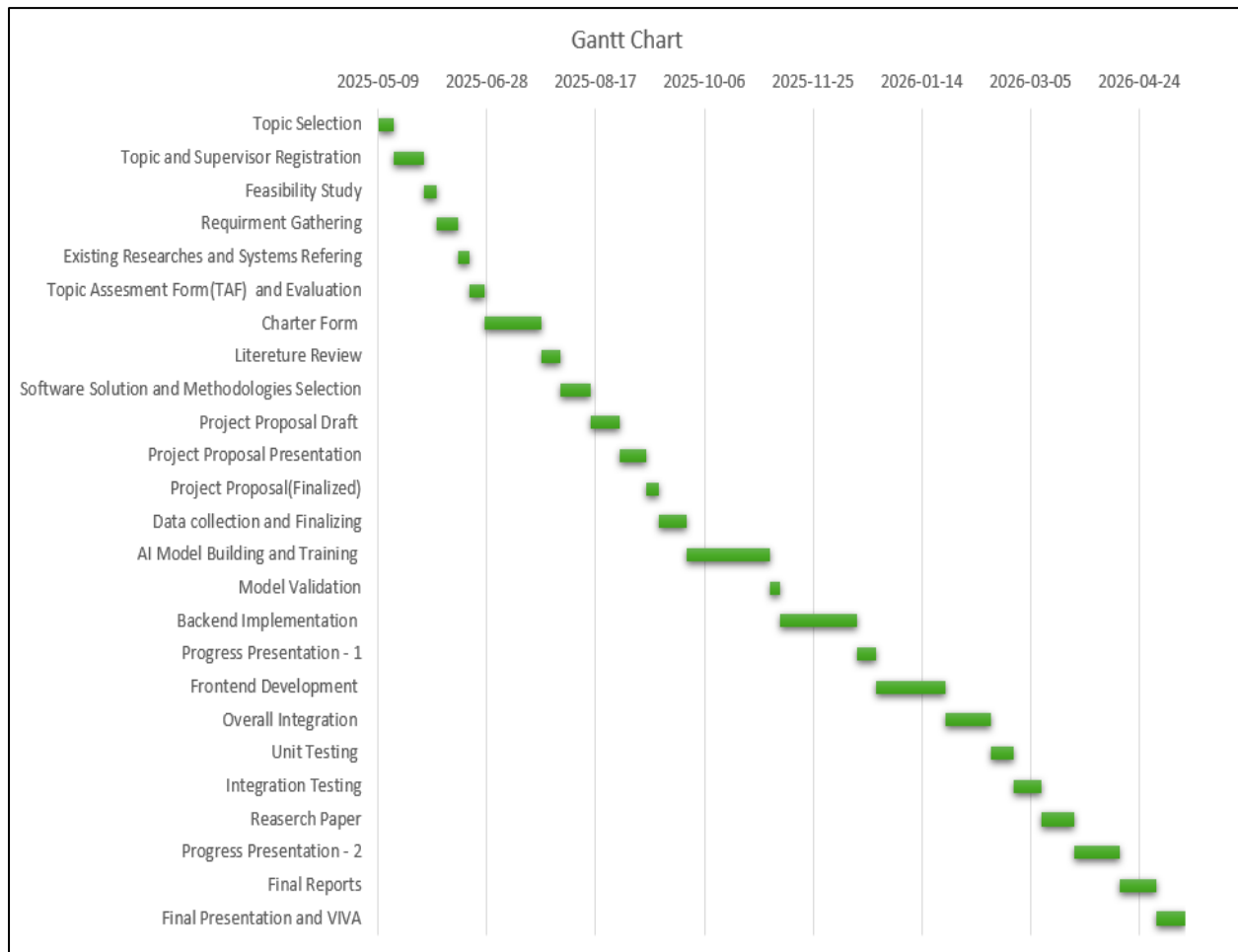


Figure 10.1: Gantt chart

The Gantt chart timeline goes from May 2025 to May 2026 and shows all the important stages of the project. The first part of the work starts in May 2025 with topic selection, supervisor registration, feasibility study, and requirement gathering. After that, the project moves into reviewing existing research, completing the Topic Assessment Form, and preparing the charter. The literature review and methodology selection happen in mid-2025, followed by the project proposal, its presentation, and final submission by September 2025. From late September, the project focuses on technical development starting with data collection, AI model building, training, and validation. Then backend and frontend development are carried out along with integration work. Early 2026 is reserved for testing activities such as unit testing and integration testing. The last stage of the project includes preparing the research paper, final report, and presentations. The project ends with the final presentation and viva in May 2026.

## 11. CONCLUSION

The research on the Fruit Quality Verification and Complaint Validation System shows how technology can be used to solve real-world problems in the food industry. At present, fruit quality checking is mostly done by manual inspection, which can be slow, subjective, and sometimes unfair to both customers and sellers. Complaints raised by customers are also difficult to validate because there is no proper evidence-based process to prove whether the product was actually damaged or of poor quality.

With this system, machine learning and image analysis are applied to identify the freshness, ripeness, and defects of fruits in an objective and reliable way. The automated checking reduces human errors and ensures a more standard method of evaluating fruit quality. At the same time, the complaint validation feature provides a fair process where uploaded images can be analyzed to confirm the validity of complaints. This not only protects customers from receiving low-quality products but also prevents sellers from false or unfair claims.

The system has the potential to benefit farmers, distributors, online fruit sellers, and customers by increasing trust in the supply chain. It also introduces transparency and efficiency in the way complaints are handled, leading to improved satisfaction for both parties. Even though this work is still at a development stage, it provides a solid base for future improvements like expanding to more fruit types, integrating IoT sensors, or linking with large-scale e-commerce platforms. Overall, the project shows a practical and scalable solution to improve fruit quality management and complaint handling.

## 12. REFERENCES

- [1] R. P. A. & R. H. Wimalaratana, "Reducing post-harvest fruit loss in Sri Lanka's supply chains," *Sri Lanka Journal of Agriculture*, vol. 12(2), p. 45–57.
- [2] J. M. M. H.-G. R. e. a. Naranjo-Torres, "A review of deep learning in agriculture: Techniques and applications," *Computers and Electronics in Agriculture*, p. 178, 2020.
- [3] Y. C. Y. & W. J. Li, "Apple defect detection using deep convolutional networks," *Computers and Electronics in Agriculture*, p. 167, 2019.
- [4] C. W. T. & F. K. Jayasundara, "Traceability systems for Sri Lanka's fruit exports: Challenges and opportunities," *Tropical Agricultural Research*, vol. 32, p. 67–78, 2021.
- [5] K. S. R. & S. V. Patel, "Mango ripeness classification using GLCM and SVM," *International Journal of Computer Applications*, vol. 182(5), p. 34–39, 2018.
- [6] P. H. D. & F. S. Silva, "Texture-based pineapple spoilage detection using Random Forests," *Sri Lanka Journal of Food and Agriculture*, vol. 6(1), p. 55–62, 2020.
- [7] S. R. P. & S. D. Perera, "Complaint validation using metadata in e-commerce platforms," *Journal of Information Systems Research*, vol. 9(1), p. 22–34, 2021.
- [8] A. K. V. & S. R. (. Singh, "Securing mobile applications using in-app camera enforcement," *International Journal of Computer Applications*, vol. 184(36), p. 19–25, 2022.
- [9] C. Zauner, "Implementation and benchmarking of perceptual image hash functions," 2010.

## 13. APPENDICES

### Mango Varieties Classification and Grading

[Data Card](#) [Code \(7\)](#) [Discussion \(0\)](#) [Suggestions \(0\)](#)

#### Dataset (2 directories)



Classification\_dataset  
8 directories



Grading\_dataset  
3 directories

*Figure 13.1: Mongo dataset*